

HOME SIEM LAB

Build Guide & Technical Documentation

Splunk • Sysmon • Atomic Red Team • MITRE ATT&CK

Field	Details
Author	Feroz Khan
Version	1.0
Date	April 2026
Platform	Windows 10 Pro + VirtualBox
SIEM	Splunk Enterprise 10.2.2 (Free)
Log Agent	Sysmon v15 + Splunk Universal Forwarder
Attack Simulator	Atomic Red Team (invoke-atomicredteam)
GitHub	github.com/ferozahmadkhan
Difficulty	Beginner-Intermediate
Time to Build	10-12hours

1. Project Overview

This project documents the end-to-end build of a Home SIEM (Security Information and Event Management) Lab designed to develop real-world SOC analyst skills. The lab simulates a corporate blue team environment on a single home machine using free and open-source tools.

The lab was built as Tier 1 of a structured SOC analyst skill-building curriculum, covering log ingestion, threat detection, attack simulation, dashboard creation, and incident response documentation.

1.1 What You Will Build

By following this guide, you will have a fully functional SIEM pipeline that:

- Ingests real Windows security logs and Sysmon telemetry from a virtual machine
- Ships logs to Splunk for storage, searching, and analysis
- Detects 5 real attack techniques mapped to MITRE ATT&CK
- Visualises attack activity on a live SOC dashboard
- Produces professional Incident Response documentation

1.2 Architecture Diagram

Layer	Component	Role
Host Machine	Splunk Enterprise (Free)	SIEM — receives, indexes and searches all logs
Virtual Machine	Windows 10 Pro VM (VirtualBox)	Victim endpoint — generates security logs
Log Agent	Sysmon v15 + SwiftOnSecurity config	Deep telemetry — process, network, file events
Log Shipper	Splunk Universal Forwarder	Ships logs from VM to Splunk on port 9997
Attack Simulator	Atomic Red Team	Simulates real MITRE ATT&CK techniques
Detection	Custom SPL queries	Identifies malicious patterns in the log data
Visualisation	Splunk Dashboard	Single-pane SOC view of all detections

1.3 Skills Developed

- SIEM administration and configuration (Splunk)
- Log ingestion pipeline design (Forwarder → Receiver → Index)
- Windows Event Log analysis (Security, System, Application)
- Sysmon deployment and tuning
- SPL (Splunk Processing Language) query writing
- MITRE ATT&CK framework — practical technique mapping
- Attack simulation using Atomic Red Team
- SOC dashboard design and visualisation
- Incident Response report writing

2. Prerequisites

2.1 Hardware Requirements

Component	Minimum	Recommended
RAM	8 GB	16 GB
CPU	4 cores	6+ cores
Disk Space	80 GB free	120 GB free
OS	Windows 10/11 (64-bit)	Windows 10/11 (64-bit)
Internet	Required for downloads	Required for downloads

2.2 Software Downloads

Tool	Where to Download	Cost
VirtualBox	virtualbox.org/wiki/Downloads	Free
VirtualBox Extension Pack	virtualbox.org/wiki/Downloads	Free
Windows 10 Enterprise ISO	microsoft.com/en-us/evalcenter	Free (90-day eval)
Splunk Enterprise	splunk.com/en_us/download/splunk-enterprise.html	Free (60-day)
Splunk Universal Forwarder	splunk.com/en_us/download/universal-forwarder.html	Free
Sysmon	learn.microsoft.com/sysinternals/downloads/sysmon	Free
SwiftOnSecurity Sysmon Config	github.com/SwiftOnSecurity/sysmon-config	Free
Atomic Red Team	github.com/redcanaryco/invoke-atomicredteam	Free

3. Phase 1 — Environment Setup

In this phase you build the isolated virtual machine that acts as your monitored victim endpoint. Keeping the victim machine isolated in a VM protects your real PC from any attack simulation tools.

3.1 Install VirtualBox

1. Download the Windows hosts installer from [virtualbox.org](https://www.virtualbox.org)
2. Download the VirtualBox Extension Pack (same version)
3. Run the main installer — click through all defaults, accept the network warning
4. Double-click the Extension Pack file — VirtualBox opens and installs it automatically

Tip: The network warning during installation briefly disconnects your internet for 2-3 seconds. This is normal — VirtualBox is installing a virtual network adapter.

3.2 Download Windows 10 ISO

5. Go to microsoft.com/en-us/evalcenter/evaluate-windows-10-enterprise
6. Fill in the short form (name, email, company) — any values are accepted
7. Select 64-bit and download — file is approximately 4-5 GB

This is a 90-day evaluation copy — completely free and legal for lab use.

3.3 Create the Windows 10 Virtual Machine

8. Open VirtualBox → click New
9. Set Name: Windows10-Victim, Type: Microsoft Windows, Version: Windows 10 (64-bit)
10. Select your downloaded ISO under ISO Image
11. Check Skip Unattended Installation — click Next
12. Set RAM to 4096 MB minimum (6144 MB recommended) — click Next
13. Set disk size to 60 GB — click Next → Finish

Important: Tick 'Skip Unattended Installation' so you control the Windows setup process yourself.

3.4 Install Windows 10

14. Start the VM — Windows Setup boots from the ISO
15. Select English → Install Now → I don't have a product key → Windows 10 Enterprise
16. Select Custom Install → select the 60 GB virtual disk → Next
17. Wait 10-20 minutes for installation — VM reboots automatically
18. Region: India → Keyboard: US → Skip second keyboard

19. Click Domain join instead (bottom-left) — creates a local account without Microsoft account
20. Username: victim → leave password blank → complete setup
21. Turn off all privacy settings → skip Cortana

3.5 Install Guest Additions

Guest Additions makes the VM window resizable, fixes mouse trapping, and enables clipboard sharing between host and VM.

22. With the VM running → top menu: Devices → Insert Guest Additions CD Image
23. Inside VM → File Explorer → This PC → open the VirtualBox Guest Additions drive
24. Run VBoxWindowsAdditions.exe → Next → Next → Install → Reboot Now

3.6 Take a Snapshot

Snapshots are save points. If anything breaks later, restore this snapshot and start fresh.

25. VM running → VirtualBox top menu: Machine → Take Snapshot
26. Name: Clean-Install, Description: Fresh Windows 10 with Guest Additions

4. Phase 2 — Log Ingestion Pipeline

This phase builds the complete pipeline that moves logs from your VM into Splunk. Think of it as installing cameras (Sysmon), cables (Universal Forwarder), and a recording system (Splunk).

4.1 Install Splunk Enterprise (Host PC)

27. Download Splunk Enterprise (Windows .msi) from splunk.com — create a free account
28. Run the installer → accept license → choose local system account → set admin password
29. Splunk starts automatically — open <http://localhost:8000> in your browser
30. Log in with the admin credentials you just set

4.2 Configure Receiving Port

This opens the door in Splunk so the Forwarder can deliver logs.

31. Splunk → Settings → Forwarding and Receiving → Configure Receiving
32. Click New Receiving Port → enter 9997 → Save

4.3 Create Splunk Indexes

Indexes are storage buckets. We separate Windows logs and Sysmon logs for easier searching.

33. Splunk → Settings → Indexes → New Index
34. Create index name: windows → Save
35. Repeat: create index name: sysmon → Save

4.4 Find Your Host PC IP Address

The Forwarder on the VM needs to know where to send logs — your host PC's IP address.

36. On your real PC: press Windows Key + R → type cmd → press Enter
37. Type ipconfig → look for VirtualBox Host-Only Network
38. Note the IPv4 address (typically 192.168.xx.x)

4.5 Install Splunk Universal Forwarder (VM)

39. Inside the VM → download Splunk Universal Forwarder (Windows 64-bit .msi)
40. Run installer → accept license → select On-premises Splunk Enterprise instance
41. Set username/password for the forwarder
42. Leave Deployment Server blank → click Next
43. Receiving Indexer: enter your host PC IP and port 9997 → Install

4.6 Configure inputs.conf

This file tells the Forwarder exactly which logs to collect. Save it to:

`C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf`

Open Notepad as Administrator before editing this file — standard Notepad cannot save to this protected directory.

```
[WinEventLog://Security]
disabled = 0
index = windows
renderXml = false
```

```
[WinEventLog://System]
disabled = 0
index = windows
renderXml = false

[WinEventLog://Application]
disabled = 0
index = windows
renderXml = false

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
index = sysmon
renderXml = false
```

4.7 Install Sysmon

Sysmon provides deep visibility into process creation, network connections, file creation, and registry changes — far beyond what standard Windows logs offer.

44. Inside VM → download Sysmon from Microsoft Sysinternals
45. Download the SwiftOnSecurity config from github.com/SwiftOnSecurity/sysmon-config
46. Extract both to C:\Sysmon\
47. Open CMD as Administrator → navigate to C:\Sysmon\
48. Run the install command:

```
sysmon64.exe -accepteula -i sysmonconfig-export.xml
```

49. Verify Sysmon is running: services.msc → find Sysmon64 → Status: Running

4.8 Fix Forwarder Service Account

Sysmon logs are protected — the Forwarder must run as SYSTEM to read them.

50. services.msc → find SplunkForwarder → right-click → Properties
51. Click the Log On tab → select Local System account → Apply → OK
52. Right-click SplunkForwarder → Restart

This is a critical step. Without SYSTEM permissions the Forwarder cannot read Sysmon's Operational log channel and the sysmon index will remain empty.

4.9 Verify the Pipeline

Check that logs are flowing in Splunk:

```
index=windows    → should return Windows Event Log events
index=sysmon     → should return Sysmon process/network events
```

Take a snapshot once both indexes are returning results:

53. Machine → Take Snapshot → Name: Sysmon-Working

5. Phase 3 — Attack Simulation & Detection

This phase uses Atomic Red Team to simulate real MITRE ATT&CK techniques on the VM, then detects them using custom SPL queries in Splunk.

5.1 Install Atomic Red Team

Atomic Red Team is a library of small, focused attack simulations each mapped to a MITRE ATT&CK technique. Run all commands in PowerShell as Administrator inside the VM.

54. Allow PowerShell script execution:

```
Set-ExecutionPolicy Bypass -Scope CurrentUser -Force
```

55. Install NuGet provider:

```
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Force
```

56. Install powershell-yaml dependency:

```
Install-Module -Name powershell-yaml -Force -AllowClobber
```

57. Install Atomic Red Team:

```
IEX (IWR 'https://raw.githubusercontent.com/redcanaryco/atomicredteam-
atomicredteam/master/install-atomicredteam.ps1' -UseBasicParsing)
Install-AtomicRedTeam -getAtomics -Force
```

58. Import the module (do this at the start of every PowerShell session):

```
Import-Module "C:\AtomicRedTeam\invoke-atomicredteam\Invoke-AtomicRedTeam.ps1" -
Force
```

5.2 Detection 1 — Account Discovery (T1087.001)

What it simulates

Post-exploitation reconnaissance — an attacker who just gained access runs commands to discover what accounts and groups exist on the machine.

Run the simulation

```
Invoke-AtomicTest T1087.001
```

SPL Detection Query

```
index=sysmon EventCode=1
| search CommandLine="*net user*" OR CommandLine="*whoami*" OR CommandLine="*net localgroup*"
| table _time, host, User, CommandLine, ParentCommandLine
| sort - _time
```

What to look for

- whoami — attacker checking their privilege level
- net user — enumerating all local user accounts
- net localgroup — listing local groups (especially Administrators)
- Parent process = powershell.exe — commands launched from PowerShell session

5.3 Detection 2 — Brute Force (T1110.001)

What it simulates

Repeated failed login attempts against local accounts using automated password guessing.

Run the simulation

```
Invoke-AtomicTest T1110.001
```

SPL Detection Query

```
index=windows EventCode=4625
| stats count by Source_Network_Address, Account_Name, host, Logon_Type
| where count > 3
| sort - count
```

Key Windows Event Fields

Field Name	Meaning
Account_Name	The account being attacked
Source_Network_Address	Where the attack is coming from
Logon_Type	How login was attempted (2=local, 3=network, 10=RDP)
Failure_Reason	Why it failed (wrong password, locked account, etc.)

5.4 Detection 3 — Suspicious PowerShell (T1059.001)

What it simulates

Malicious PowerShell execution using attacker tradecraft — encoded commands, download cradles, execution policy bypasses, and in-memory offensive tools.

Run the simulation

```
Invoke-AtomicTest T1059.001
```

SPL Detection Query

```
index=sysmon EventCode=1 Image="*powershell.exe"
| eval suspicious=if(
    match(CommandLine, "(?i) (-enc|-EncodedCommand|-ExecutionPolicy Bypass|-
    WindowStyle Hidden|-nop|-noni|IEX|Invoke-Expression|DownloadString|WebClient)"),
    "YES", "NO")
| where suspicious="YES"
| table _time, host, User, CommandLine, ParentImage, suspicious
| sort - _time
```

PowerShell Red Flags

Flag	Meaning	Severity
-enc / -EncodedCommand	Command is base64 encoded to hide content	High
-WindowStyle Hidden	Running invisibly — user cannot see window	High
-ExecutionPolicy Bypass	Bypassing Windows security controls	High
-nop (NoProfile)	Skipping normal PowerShell profile to run faster/stealthier	Medium
IEX / Invoke-Expression	Executing downloaded or dynamically generated code	Critical
DownloadString / WebClient	Downloading code from the internet to execute	Critical

5.5 Detection 4 — Backdoor Admin Account (T1136.001)

What it simulates

An attacker creates hidden local admin accounts to maintain persistent access even if the original compromised account is discovered.

Run the simulation

```
Invoke-AtomicTest T1136.001
```

SPL Detection Query

```
index=windows (EventCode=4720 OR EventCode=4732)
| eval action=case(
    EventCode=4720, "New Account Created",
    EventCode=4732, "Added to Administrators Group")
```

```
| table _time, host, Account_Name, action, src_user
| sort - _time
```

What to look for

- EventCode 4720 (new account created) immediately followed by EventCode 4732 (added to admins)
- Both events triggered by a non-IT standard user account
- Multiple accounts created in rapid succession (seconds apart)
- Account names that look programmatic: T1136.001_CMD, NewLocalUser, backdoor

5.6 Detection 5 — Credential Dumping / Mimikatz (T1003.001)

What it simulates

Multiple LSASS (Local Security Authority Subsystem Service) credential dumping techniques including Mimikatz, ProcDump, rundll32 comsvcs.dll abuse, and xordump.

Run the simulation

```
Invoke-AtomicTest T1003.001
```

SPL Detection Query

```
index=sysmon EventCode=1
| search CommandLine="*lsass*" OR CommandLine="*mimikatz*" OR
CommandLine="*sekurlsa*" OR CommandLine="*procdump*"
| table _time, host, User, CommandLine, ParentCommandLine
| sort - _time
```

Credential Dumping Techniques Detected

Tool / Technique	Command Pattern	Evasion Method
Mimikatz	mimikatz.exe sekurlsa::logonpasswords	Direct execution
Invoke-Mimikatz	IEX DownloadString(...Invoke-Mimikatz.ps1)	In-memory, no disk artifact
ProcDump	procdump.exe -mm lsass.exe lsass_dump.dmp	Legitimate Microsoft tool abused
comsvcs.dll	rundll32.exe comsvcs.dll MiniDump [PID]	LOLBin — Living off the Land
xordump	xordump.exe -out lsass.dmp -x 0x41	XOR-encrypted dump to evade AV

6. Phase 4 — SOC Dashboard

The dashboard provides a single-pane view of all detection activity. This is the primary interface real SOC analysts use to monitor their environment.

6.1 Create the Dashboard

59. Splunk → Dashboards → Create New Dashboard

60. Title: SOC Detection Lab, select Classic Dashboard → Create

6.2 Dashboard Panels

Panel	Type	What it Shows
Attack Overview Counter	Single Value	Total threat events detected in last 24 hours
Top Targeted Accounts	Bar Chart	Which accounts received the most brute force attempts
Suspicious PowerShell Timeline	Line Chart	PowerShell attack events plotted over time
Backdoor Account Activity	Table	New accounts created and admin group additions
Credential Dumping Attempts	Table	All LSASS access and Mimikatz-related commands
Overall Attack Timeline	Area Chart	All attack events plotted over time — shows attack spikes

6.3 Panel Queries

Panel 1 — Attack Counter

```
index=sysmon OR index=windows
(EventCode=4625 OR EventCode=4720 OR EventCode=4732
OR (EventCode=1 AND (CommandLine="*mimikatz*" OR CommandLine="*lsass*")))
OR (EventCode=1 AND CommandLine="*net user*")
| stats count as "Total Threats Detected"
```

Panel 2 — Brute Force Bar Chart

```
index=windows EventCode=4625
| stats count by Account_Name
| sort - count | head 10
| rename Account_Name as "Targeted Account", count as "Failed Attempts"
```

Panel 3 — PowerShell Timeline

```
index=sysmon EventCode=1 Image="*powershell.exe"
```

```
| search CommandLine="*-enc*" OR CommandLine="*bypass*" OR CommandLine="*IEX*"
| timechart count span=1h as "Suspicious PowerShell Events"
```

Panel 6 — Overall Attack Timeline

```
index=sysmon OR index=windows
(EventCode=4625 OR EventCode=4720 OR EventCode=4732
OR (EventCode=1 AND (CommandLine="*mimikatz*" OR CommandLine="*net user*" OR
CommandLine="*-enc*")))
| timechart count span=30m as "Attack Events"
```

7. Key Windows Event IDs Reference

Event ID	Log	Meaning	Why It Matters
4624	Security	Successful logon	Baseline — track normal login patterns
4625	Security	Failed logon	Brute force detection — high count = attack
4648	Security	Logon using explicit credentials	Pass-the-hash, lateral movement
4672	Security	Special privileges assigned	Admin logon — watch for unexpected accounts
4720	Security	User account created	Backdoor account creation
4732	Security	User added to security group	Privilege escalation — new admin
4776	Security	NTLM authentication	Credential validation — watch for failures
1 (Sysmon)	Sysmon	Process created	Everything that launches — core detection source
3 (Sysmon)	Sysmon	Network connection	C2 beaconing, data exfiltration
10 (Sysmon)	Sysmon	Process accessed another	LSASS access = credential dumping
11 (Sysmon)	Sysmon	File created	Malware dropping files, ransomware
13 (Sysmon)	Sysmon	Registry value set	Persistence mechanisms, malware config

8. MITRE ATT&CK Technique Mapping

Technique ID	Name	Tactic	Detection Method	Event Source
T1087.001	Account Discovery: Local Account	Discovery	CommandLine contains net user / whoami	Sysmon EID 1
T1110.001	Brute Force: Password Guessing	Credential Access	EventCode 4625 count > threshold	Windows Security
T1059.001	Command & Scripting: PowerShell	Execution	Suspicious flags in CommandLine	Sysmon EID 1
T1136.001	Create Account: Local Account	Persistence	EventCode 4720 + 4732 correlation	Windows Security
T1003.001	OS Cred Dumping: LSASS Memory	Credential Access	CommandLine contains lsass / mimikatz	Sysmon EID 1

9. Troubleshooting Guide

Problem	Cause	Fix
No driver found during Windows install	VirtualBox using NVMe controller	Settings → Storage → change controller from NVMe to SATA (AHCI)
index=windows returns no results	Index not created in Splunk	Settings → Indexes → New Index → name: windows
index=sysmon returns no results	Forwarder running as wrong account	services.msc → SplunkForwarder → Log On → Local System account → Restart
Splunk port 8000 not loading	Splunk service not running	services.msc → SplunkD → Start
Forwarder not sending logs	inputs.conf not saved correctly	Open Notepad as Administrator → save inputs.conf to correct path → restart forwarder
Sysmon events missing EventCode 10	Sysmon config not capturing process access	Update sysmonconfig-export.xml to enable ProcessAccess events
Atomic Red Team install fails	powershell-yaml module missing	Run: Install-Module powershell-yaml -Force -AllowClobber
VM window too small	Guest Additions not installed	Devices → Insert Guest Additions CD → run VBoxWindowsAdditions.exe

10. About This Project

Field	Details
Author	Feroz Khan
GitHub	github.com/ferozahmadkhan
Project Type	SOC Analyst Home Lab — Tier 1
Tools Used	Splunk Enterprise, Sysmon, Splunk Universal Forwarder, Atomic Red Team, VirtualBox
Framework	MITRE ATT&CK Enterprise
Techniques Covered	T1087.001, T1110.001, T1059.001, T1136.001, T1003.001
Total Detections Built	5 custom SPL detection rules
Dashboard Panels	6 panels covering all detection categories
Date Completed	April 2026
License	MIT — free to use, modify, and share with attribution

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